

Executive Summary: This week after our team suffered the largest hurdle of last week's microphone part, we made the most amount of combined progress. The hardware side of our WAD has finished the first prototype on breadboards and will begin work on the final prototype on protoboards for the Project Manager to finish in a casing. Once the final prototype is finished the hardware team plans on finishing all necessary documentation and future planning with a PCB design. The software team (shoutout solo Alex Ho) got the furthest in solving the gun shot detection problem then we have so far. Finishing the frequency analysis portion of the software for the audio signal processing, our team is ready for all testing portions and data gathering for our final presentation. For the overall documentation side of things, our team knows the assignments given to them and we are confident that within the remaining two weeks everything will be accomplished on time if not early!

Hardware:

Cody C.

Progress:

- 4/13 – Hardware building with Ryan (2 Hours 30 Minutes)
 - Solder pins to x2 Boost Converter
 - Fix soldering pins for Microphone chip
- Finish up wiring for full hardware system on breadboard (~30 Minutes)
- 4/15 – Team meeting (~3 Hours)
 - Test full hardware system with hardware system
 - Microphone hardware works outside plugged into laptop

- Having issues with the battery power. The boost converter is outputting 6.7V with current of 200mA but it does not power the whole hardware system
 - Build full hardware system onto smaller breadboard
- 4/17 – Troubleshoot hardware with Ryan (~2 Hours)
 - Conducted multiple testing of the boost converter
 - Added capacitors between Vin and GND to limit the noise
 - Added another capacitor between boost converter and Vin
 - Added diode between output of boost converter to Vin
 - Measured all the testing above with 5V
 - Direct connection of 3.7V to Arduino Nano ESP 32 works and powers both GPS and microphone with Bluetooth
 - Conducted field testing to make sure it works outside
 - Received signal walking around outside with 3.7V battery
 - Direct connection with DC power supply stepping down
 - Found out that the whole system will run all the way down to 2.8V before everything turns off.

Total Hours Worked: ~ 8 Hours

Future Work/Plan

- Figure out how long the run time?
- Update GitHub
- Start Protoboard

- Research how to create a PCB for the prototype to have 3D model to show
- Start/Create Hardware Documentation

Hardware:

Ryan K.

Progress:

- 4/13 - Constructing Hardware (~3hrs)
 - Solder components that needed it
 - Wired and troubleshoot complete device
- 4/15 - Team Meeting (~2hrs)
 - Tested all components of the device
 - Could not get the device to work independently
- 4/17 - Hardware Troubleshooting (~2hrs)
 - Troubleshoot boost converter to get the device to work independently
 - Tested device without boost converter, which causes everything to work properly
 - Tested different voltage values to the Arduino to see when a direct connection from the battery would become obsolete

Total Hours Worked: ~7 Hours

Future Work/Plan:

- Construct Protoboard
- Begin writing required hardware documentation

Software:

Alex H.

Progress:

- 4/14 (2 h 7 min)
 - Readjusted internal settings inside of the software to configure the GPS updates to occur approximately every 1 second
 - Developed new software that accurately lowers the audio dB levels to more realistic readings.
 - Collected readings from the analog microphone to determine the average noise floor of the microphone. Compared readings from the Wearable Alert Device with a sound meter mobile app. The average noise floor appeared to be about 15 dB.
 - New software subtracts about 15 dB from the raw dB values to filter out the noise floor. Made this subtraction value easily customizable. If future developers want to use a different value instead of 15 dB, they just need to change a single number in the config.h file.
 - Combined peak detection software with Bluetooth software so that whenever a dB reading exceeds the customizable threshold, it will automatically transmit the data wirelessly without the need to press a button.
- 4/15 (1 h 58 min)
 - Tested new audio software with the team. Other team members seem to be satisfied with the current state of the audio software.
 - Performed numerous tests with the peak detection, adjusting the gunshot dB threshold level to see what would be a reasonable threshold to set the device to for the final demonstration.

- Troubleshooted boost converter issues with the hardware portion of the device.
- 4/19 (6 h 14 min)
 - Developed frequency analysis portion of the software for the audio signal processing.
 - Applies a Hamming window function to the voltages output by the microphone.
 - Performs the Fast Fourier Transform on the voltages to obtain a list of the strongest energy magnitudes in the audio signal and their associated frequencies.
 - Based on research, the muzzle blast from gunshots have the most energy below 500 Hz, so the software checks if the highest magnitude of energy is somewhere near this 500 Hz value by a certain margin. This margin is also easily customizable by future software developers, as only a single number needs to be changed to adjust it.
 - Started writing the user manual for the Wearable Alert Device.
 - Started writing the README file for the Wearable Alert Device on the GitHub project repository, which will eventually be converted into a standard doc/pdf file to become the project's software documentation.

Total Hours Worked: 10 h 19 min

Future Work/Plan:

- Continue writing user manual
- Continue writing README/software documentation

- Develop some form of methodology of proper, valid data collection so that we can present some solid data at the final demonstration.

Project Manager:

Jonas W.

Progress:

- 4/15 – Team meeting (~2 Hours)
 - Test full software system with hardware
 - Found issue of the board not working independently with the battery pack.
 - Scheduled plans between hardware group to resolve issue and find game plan
- 4/16 PM Work (1 Hour)
 - Reschedule the Final Demo Presentation
 - Work on the BOM
- 4/17 Potential Casing Work (1 Hour)
 - Research on how and where to build case for the device once Hardware is finished on protoboard
 - Research on which modeling software to use and if it would be worth the time this semester to have ready for next semester's group
 - Re did OU Fabrication Lab certifications so I could have access to 3D printing at their facilities
- 4/18 PM Purchases/ PM Work (1 Hour)
 - Purchased binders for final demo booklet
 - Purchased paper for the 11x7 schematics to be printed on
 - Worked on outline of Final DEMO booklet

Total Hours Worked: ~5 Hours

Future Work/Plan:

- Continue work on the User Manual help
- Continue work on formatting and helping with hardware design document
- Continue work on BOM and Final Demo Booklet/ Slides
- Continue work on Casing for final prototype
- Reach out to when we can go test the gun shot detection for possibly week on 27th